

Proposal to Replace Incandescent Exit Signs with Self-Illuminating Exit Signs

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Many facilities rely on traditional incandescent exit signs, which require continuous electricity and frequent maintenance. These systems can be inefficient, costly over time, and vulnerable to power outages. Based on recent data from the Marine Corps Development and Education Center (MCDEC), replacing incandescent exit signs with self-illuminating exit signs can significantly reduce long-term costs and improve reliability (Kolin, 2022). This proposal recommends upgrading existing exit signage to self-illuminating systems to improve efficiency, reduce maintenance, and enhance safety.

Traditional exit signs rely on electrical power and require ongoing maintenance like replacing bulbs and overall daily energy costs. Over time, these expenses add up and create unnecessary operational costs for companies. Also, during power failures or electrical disturbances, these traditional exit signs may fail unless there are backup systems in place. This creates a potential safety risk, especially in emergency situations where clear exit visibility is critical.

This proposal recommends replacing all existing incandescent exit signs with self-illuminating exit signs. These signs operate without electricity and can last between 10 to 12 years without requiring maintenance. Because they do not rely on any electrical circuits, they will still function even during power outages, this is a much more reliable and safe option.

Self-illuminating exit signs are also easier to install, as they do not require complex wiring or dedicated electrical circuits. Installation costs are comparable to traditional systems, with labor estimated at approximately \$10 per sign (Kolin, 2022). This makes the transition both practical and cost-effective.

One of the primary benefits of switching to self-illuminating exit signs is the reduction in energy costs. For example, a facility with 400 incandescent exit signs operating continuously can consume over 140,000 kWh per year, resulting in significant electricity expenses (Kolin, 2022). Eliminating this energy usage can lead to substantial annual savings.

In addition to energy savings, maintenance costs are greatly reduced. Traditional exit signs require regular bulb replacements, inventory management, and labor for servicing. Self-illuminating signs eliminate these ongoing expenses.

Self-illuminating exit signs do not depend on electrical systems, which means they remain operational during power outages. This improves overall safety and ensures compliance with emergency preparedness standards.

Although the initial cost of self-illuminating signs ranges from \$200 to \$500 depending on the model, the long-term savings outweigh the upfront investment. In the MCDEC case study, the first-year savings exceeded \$37,000, with a payback period of approximately 2.6 years (Kolin, 2022).

The total cost of implementing self-illuminating exit signs will vary depending on the number of signs and the type selected. However, installation costs are similar to traditional systems due to reduced wiring requirements. Over time, the elimination of electricity usage and maintenance costs results in significant financial savings.

Additional savings may include reduced labor, fewer replacement materials, and lower operational overhead. These factors contribute to a strong return on investment and make this proposal financially realistic.

Replacing incandescent exit signs with self-illuminating exit signs is a practical and cost-effective solution to improve safety and reduce operational expenses. The benefits of energy savings, reduced maintenance, and improved reliability make this upgrade a strong investment for any facility. Based on the data provided, this proposal offers a clear opportunity to enhance efficiency while lowering long-term costs. It is recommended that the company move forward with this upgrade to improve both safety and financial performance.

References

Kolin, P. C. (2022). Self-illuminating exit signs (pp. 458–459). In *Successful writing at work* (12th ed.). Cengage Learning.